

A8: SERIES Y SUCESIONES

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1) Mostrar n términos de la sucesión siguiente:

0, 1, 0, 1, 2, 1, 0, 1, 2, 3, 2, 1, 0, 1, 2, 3, 4, 3, 2.....

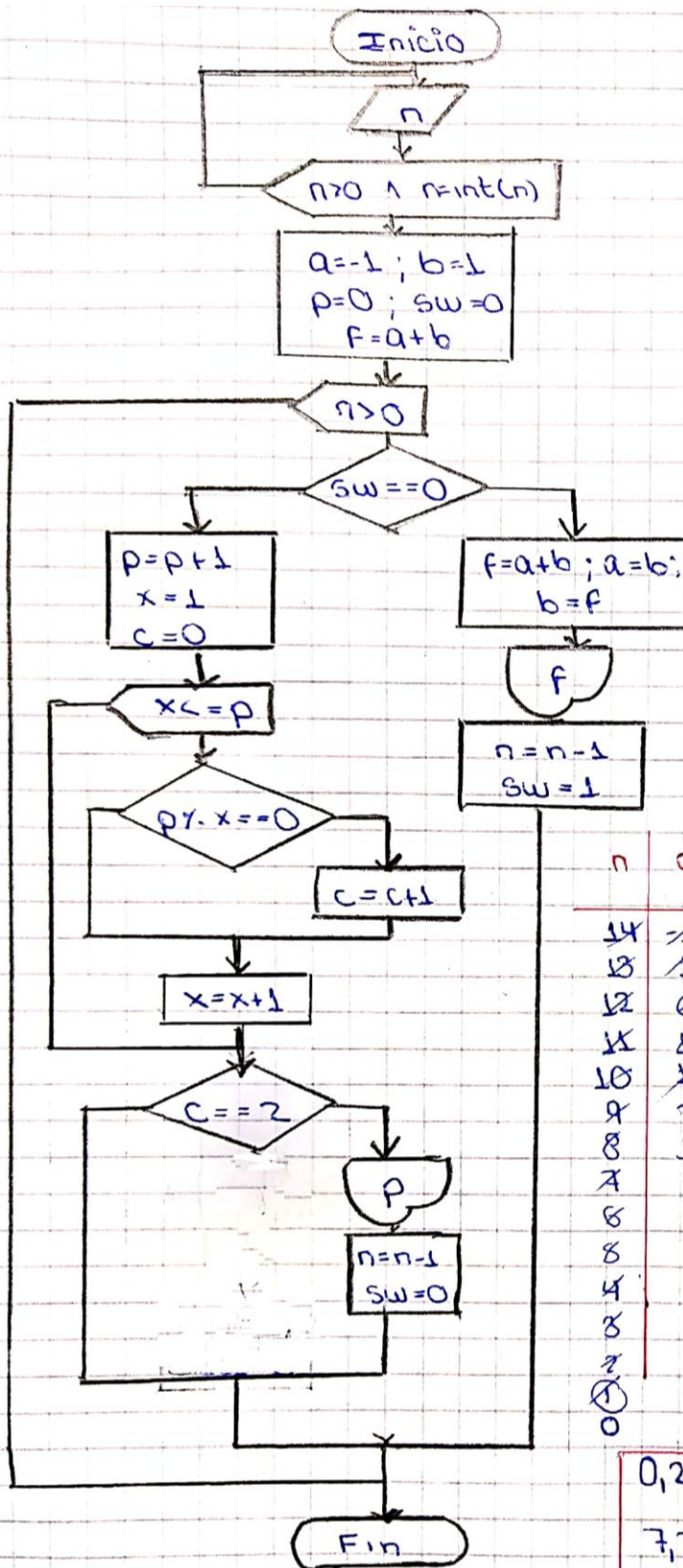
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1  #Ejercicio 1 P8
2  n=int(input("Ingresar los terminos: "))
3  g=1
4  c=-1
5  for i in range(n):
6      if c<g:
7          c=c+1
8      else:
9          c=0
10         g=g+1
11     if i+1 != n:
12         print(c, end=", ")
13     else:
14         print(c) |
```

14

0, 1, 0, 1, 2, 0, 1, 2, 3, 0, 1, 2, 3, 4

2) Mostrar n términos de la sucesión siguiente:

0, 2, 1, 3, 1, 5, 2, 7, 3, 11, 5, 13, 8, 17, 13.....



n	a	b	p	sw	f	c	x
14	1	1	0	0	0	0	1
13	1	0	1	1	1	1	2
12	0	1	2	0	1	0	1
11	1	1	3	1	2	1	2
10	1	2	4	0	3	2	3
9	2	3	5	1	5	0	1
8	3	5	6	0	8	1	2
7			7	1		0	1
6			8	0		1	2
5			9	1		0	1
4			10	0		1	2
3			11	1		0	1
2			12	0		1	2
1			13	1		0	1
0			14	0		1	2

0, 2, 1, 3, 1, 5, 2
7, 3, 11, 5, 13, 8, 17

0, 2, 1, 3, 1, 5, 2, 7, 3, 11, 5, 13, 8, 17

```

2  n=int(input("Ingresé la cantidad: "))
3  p=0
4  sw=0
5  a=-1; b=1; f=a+b;
6  while(n>0):
7      if sw==0:
8          f=a+b; a=b; b=f;
9          print(f, end=", ")
10         n=n-1
11         sw=1
12     else:
13         p=p+1
14         x=1
15         c=0
16         while(x<=p):
17             if p%x == 0:
18                 c=c+1
19                 x=x+1
20             if c==2:
21                 print(p, end=", ")
22                 n=n-1
23                 sw=0

```

Ingresé la cantidad: 12
0, 2, 1, 3, 1, 5, 2, 7, 3, 11, 5, 13,

3) Mostrar n términos de la sucesión siguiente:

1, 2, 2, 2, 3, 3, 3, 2, 3, 4, 4, 4, 4, 2, 3, 5, 5, 5, 5, 5, 5.....

```

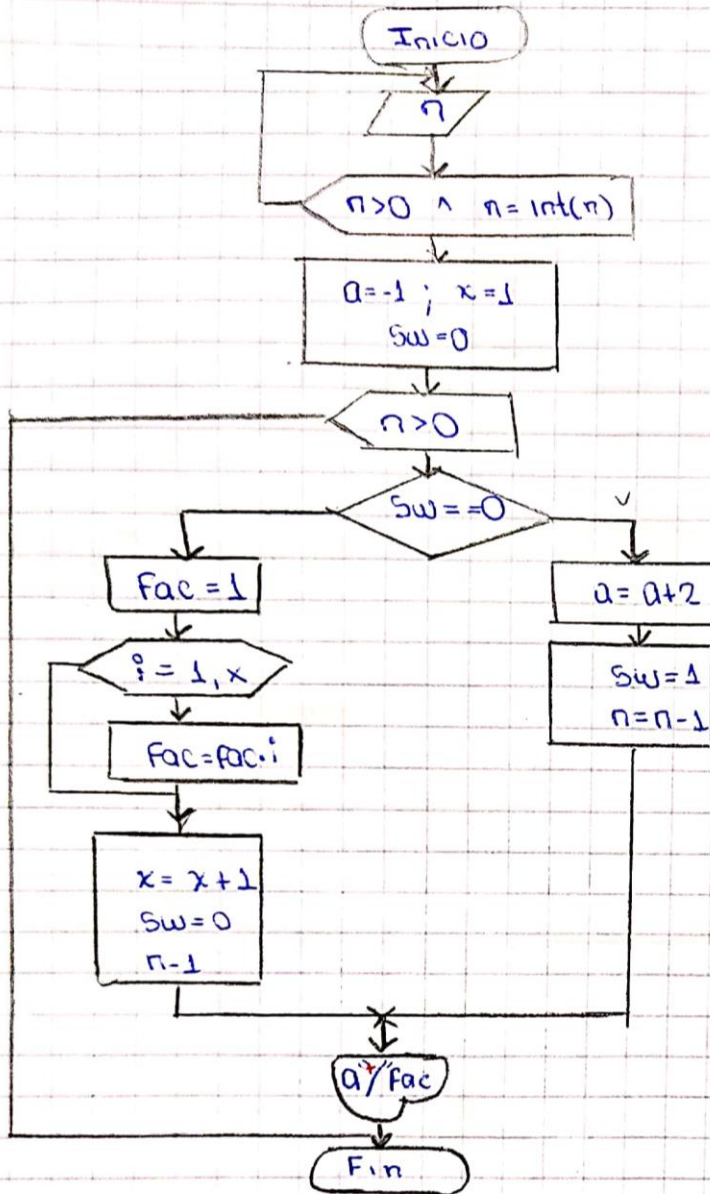
1  #Ejercicio 3 P8
2  n = int(input())
3  g = 1
4  c = 0
5  for i in range(n):
6      if c < g :
7          if i + 1 != n:
8              print(g , end = ", ")
9          else :
10             print(g)
11             c= c + 1
12         else :
13             c = 1
14             g = g + 1
15         if i + 1 != n:
16             print(g, end = ", ")

```

15
1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5

4) Mostrar n términos de la serie siguiente:

$$1/1, + 3/2 + 5/6 + 7/24 + 9/120$$



n	a	x	sw	fac
10	1	1	0	1
9	3	2	1	2
8	5	3	0	1
7	7	4	1	6
6	9	5	0	1
5	11	6	1	24
4	13	7	0	1
3	15	8	1	120
2	17	9	0	720
1	19	10	1	5040
0	19	10	0	362880
				3628800

$$\frac{1}{1} + \frac{3}{2} + \frac{5}{6} + \frac{7}{24} + \frac{9}{120}$$

$$\frac{11}{720} + \frac{13}{5040} + \frac{15}{40320} + \frac{17}{362880} + \frac{19}{3628800} + \dots$$


```

2  n=int(input("introduzca la cantidad de frac: "))
3  a=-1
4  x=1
5  sw=0
6  while(n>0):
7      if sw==0:
8          a=a+2
9          sw=1
10     else:
11         fac=1
12         for i in range(1, x+1):
13             fac=fac*i
14             x=x+1
15         print(a,"/",fac,end=" + ")
16         sw=0
17     n=n-1

```

introduzca la cantidad de frac: 6
1 / 1 + 3 / 2 + 5 / 6 + 7 / 24 + 9 / 120 + 11 / 720 +

5) Mostrar n términos de la serie siguiente:

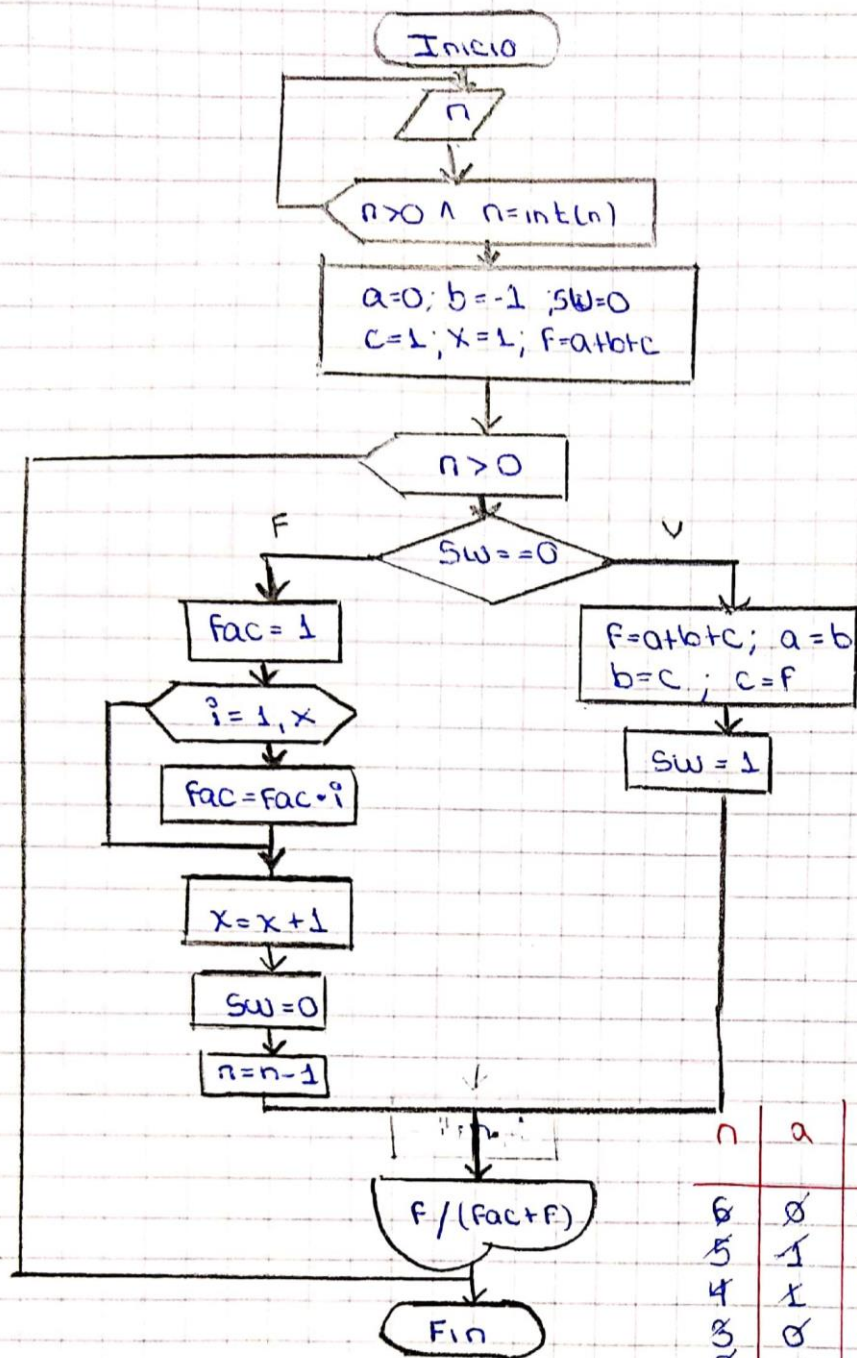
$$0/1 - 1/3 + 1/7 - 2/26 + 4/124$$

```

1  #Ejercicio 5 P8
2  n=int(input("introduzca la cantidad de frac: "))
3  a=0
4  b=-1
5  x=1
6  sw=0
7  c=1
8  n>0
9  while(n>0):
10     if sw==0:
11         f=a+b+c; a=b; b=c; c=f
12         sw=1
13     else:
14         fac=1
15         for i in range(1, x+1):
16             fac=fac*i
17             x=x+1
18         print(f, "/", fac+f, end=" , ")
19         if n%2 ==0:
20             print(end=" + ")
21         else:
22             print(end=" - ")
23         sw=0
24     n=n-1

```

introduzca la cantidad de frac: 7
0 / 1, - 0 / 2, + 1 / 7, - 1 / 25, + 2 / 122, - 4 / 724, + 7 / 5047, -



n	a	b	sw	c	x	f	fac
6	0	-1	0	1	1	0	1
5	1	1	1	0	2	0	2
4	1	0	0	0	3	1	6
3	0	0	1	1	4	1	1
2	0	1	0	1	5	2	24
1	1	1	1	2	6	4	1
0	1	2	0	4	7		120
			1				1
			0				720
			1				

$$\frac{0}{1} + \frac{0}{2} + \frac{1}{4} - \frac{1}{25} + \frac{2}{121} - \frac{4}{724}$$

